

PS 3: Week 6

Quantifying Uncertainty

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Agenda

- ① Bias vs. Noise
- ② Standard Errors
- ③ P-Values
- ④ Practice Exercise

Reminders

- Project Proposal due Monday at midnight
- Midterm signup opened this morning

Gradescope Error

If you ever see this error on a gradescope submission, do not hesitate to email. It is because the autograder failed for some reason and I can have it rerun.

Autograder Results

[Results](#)[Code](#)

The autograder failed to respond in the expected amount of time. If the autograder continues to fail, please contact help@gradescope.com.

The autograder failed to execute correctly. Contact your course staff for help in debugging this issue. Make sure to include a link to this page so that they can help you most effectively.

Research Proposal Feedback

Is the research question clear?

Is it easy to tell if it is a causal/descriptive question?

Is the data you are planning to collect well suited to answer the question?

Measuring Uncertainty

Discuss:

- ① Why do we care about measuring the uncertainty of an estimate? In other words, why just the difference in means not enough to make conclusions about the effect of a treatment?
- ② What is the difference between noise and bias?

SAMPLING BIAS



"WE RECEIVED 500 RESPONSES AND
FOUND THAT PEOPLE LOVE RESPONDING
TO SURVEYS"

sketchplanations

- Bias is a systematic deviation from the truth
- You can think of bias as a consistent inaccuracy
- You've already seen bias!
- Similar to bias in a colloquial sense:
 - If the judicial system is racially biased, it will be consistently discriminatory
 - If I have a biased point of view, I will automatically come to the same conclusion regardless of the evidence you present to me
 - If we have a biased way of researching our topic of interest, we will end up at the same (wrong) conclusion

- Noise is simply the random difference between samples
- Noise is inevitable when sampling – we don't have the true population

Bias vs. Noise

Bias can also be thought of as Inaccuracy

Noise can be thought of as Imprecision



Low Bias
Low Noise



Low Bias
High Noise



High Bias
Low Noise



High Bias
High Noise

1. Standard Deviation

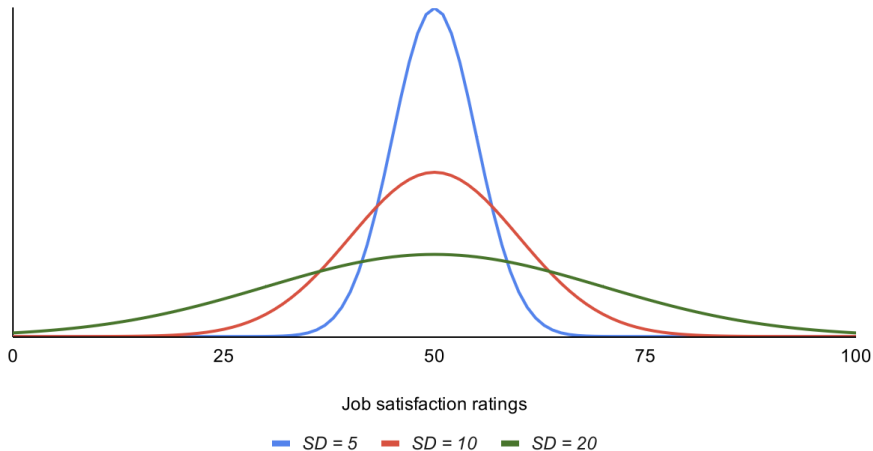
A measure of “spread” – summary of how much a set of values are away from the mean

The formula is:

$$SD = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Standard Deviation

Job satisfaction ratings of three groups



2. Standard Error

- If we took a bunch of samples and measured a value for each, the standard error is the standard deviation of those values.
- In other words, standard error is a measure of how much we'd expect our findings to vary if we could replicate our study thousands of times
- Therefore, the standard error helps us measure how much we'd expect the estimate from the experiment to differ from the truth (the real effect) by chance
- It is a measure of noise, not bias

Standard Error and other sample statistics

$$\text{Standard Error} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- n = Sample size (i.e., number of observations)
- s = Sample standard deviation
- $s_{\bar{x}}$ = Estimated standard error of the mean ($s/\sqrt{n}$)

Standard Error other sample statistics

$$\text{Standard Error} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- It can only be positive
- What do you think happens to the SE as s or n increases?

Standard Error other sample statistics

$$\text{Standard Error} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- It can only be positive
- What do you think happens to the SE as s or n increases?

As s increases, the standard error will increase as well. As the spread of the population gets bigger (standard deviation), the actual sample that we get from it will also grow.

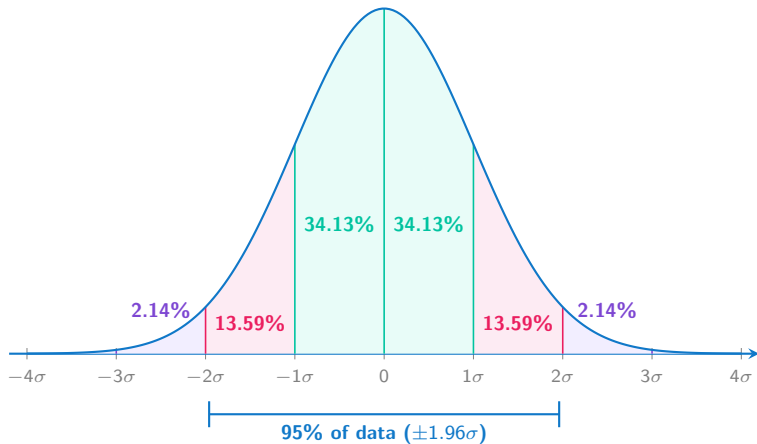
3. t-statistic

$$t = \frac{\text{Estimate}}{\text{Standard Error}}$$

- **t-statistic:** to measure how likely that an estimate of the size we saw would arise by chance, even if the treatment had no effect.
- **Estimate:** our best guess of what the true ATE is.
- If $t > 1.96$ or $t < -1.96$, the observed result is **statistically significant** (more next week).

```
#> Design: Standard
#>               Estimate Std. Error  t value  Pr(>|t|)    CI Lower  CI Upper
#> treatsociallobby 0.1161746 0.07675608 1.513555 0.1345858 -0.03687757 0.2692267
#>               DF
#> treatsociallobby 70.86971
```

t-statistic: Why 1.96?



}

Interpreting T-Statistics

```
#> Design: Standard
#>               Estimate Std. Error t value Pr(>|t|)    CI Lower CI Upper
#> treatsociallobby 0.1161746 0.07675608 1.513555 0.1345858 -0.03687757 0.2692267
#>               DF
#> treatsociallobby 70.86971
```

$$t = \frac{\text{Estimate}}{\text{Standard Error}} = \frac{0.1161746}{0.07675608} = 1.513556$$

A small estimate relative to a large standard error will result in a small t-statistic (and less confidence in the estimate being different from zero).

Attendance Survey



Hypothesis Testing

- In science we test hypotheses
 - Ex: Autocracies are more likely than democracies to go to war
- The **null hypothesis** is our default: what the world would look like if we did not observe the effect we are interested in finding
 - Ex: Autocracies and democracies are equally likely to go to war
- The **alternative hypothesis** reflects what we would observe if there was, actually, a true effect in the world
 - Ex: Autocracies and democracies are not equally likely to go to war
- We never prove one hypothesis or the other
 - Because of noise (i.e. sampling), we can never be sure we're observing the truth
 - But we can measure how likely we'd be to observe the effect we see by chance

- What's the probability that we would observe the effect we see by chance?
 - A p-value of 0.13 means that there is a 13% probability that we would observe the effect we see even if the null hypothesis was true
- We want to maximize our confidence that we are NOT observing the effect by chance
 - A p-value of greater than (or equal to) 0.05 is considered statistically insignificant
 - A p-value of less than 0.05 is statistically significant
 - We are less than 5% likely to observe this effect by chance!
 - OR: We are more than 95% likely to not be observing this effect by chance!
- This gives us evidence in support of the alternative hypothesis but does not prove it

Interpreting P-Values

Design: Standard

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
treat	0.01857536	0.01099631	1.689236	0.091361	-0.002992649	0.04014337	1669.581

- The treatment increases the outcome by about 1.9 percentage-points relative to the control.
- However, we would expect this result (or something smaller) to occur roughly 9% of the time if this effect was driven entirely by chance.

I have created a practice exercise for you to work through to give you more practice with R and a chance to determine the steps yourself and then implement it.

It is intended to be harder than the in-class assignments so don't be discouraged.

Go to the bCourses page → Section 6 → Week 6 Section Exercise

I also have created *optional* exercises that will teach you more R beyond what the class will cover. If you would prefer, you can also use this time to work on that.

Acknowledgements

Special thanks to prior GSIs, Will Halm, and Tess Louie for providing materials off of which some of these slides are based.